

N4.2.1

$$A = \begin{pmatrix} 0 & -1 & 5 \\ -1 & -3 & 5 \\ -3 & 3 & -2 \end{pmatrix}$$

$$\det(A) = a_{11} \text{Cof}(1,1) = (-1) \det(M_{11}) + a_{21} \text{Cof}(2,1) = (-1) \det(M_{21}) + a_{31} \text{Cof}(3,1) = (-1) \det(M_{31})$$

Cofactor Expansion

$\text{Cof}(i,1) = (-1)^{i+1} \det(M_{i1})$

$$a_{11} = 0 \quad M_{11} = \begin{pmatrix} -3 & 5 \\ 3 & -2 \end{pmatrix} \quad \det(M_{11}) = (-3)(-2) - (5)(3) = 6 - 15 = -9 \Rightarrow \text{Cof}(1,1) = 0$$

$$a_{21} = -1 \quad M_{21} = \begin{pmatrix} -1 & 5 \\ -3 & -2 \end{pmatrix} \quad \det(M_{21}) = (-1)(-2) - (5)(-3) = 2 + 15 = 17 \Rightarrow \text{Cof}(2,1) = -17$$

$$a_{31} = -3 \quad M_{31} = \begin{pmatrix} -1 & 5 \\ -3 & 5 \end{pmatrix} \quad \det(M_{31}) = (-1)(5) - (5)(-3) = -5 + 15 = 10 \Rightarrow \text{Cof}(3,1) = 30$$

$$\therefore 0 - 17 - 30 = -47$$

N4.2.2

When $\det(A) = 9$, find $\det(B)$

$$B = \begin{pmatrix} a_{14} & a_{12} & a_{13} & a_{11} \\ a_{24} & a_{22} & a_{23} & a_{21} \\ a_{34} & a_{32} & a_{33} & a_{31} \\ a_{44} & a_{42} & a_{43} & a_{41} \end{pmatrix}$$

Swaps: $1 \leftrightarrow 4$, $2, 3$ unchanged

$$\therefore -\det(A) \Rightarrow \det(B) = -\det(A) = -9$$

Using Theorem 3.1.2

Condition 2, $-\det(A)$

N4.2.3

$$A = \begin{pmatrix} -2 & 0 & -3 & 2 \\ 4 & 2 & 5 & -7 \\ 0 & -2 & -1 & -3 \\ 6 & 0 & 4 & 5 \\ 0 & 0 & 2 & -1 \\ 2 & 6 & 2 & 14 \\ 0 & 6 & 1 & 12 \end{pmatrix}$$

$$B = \begin{pmatrix} 2 & 0 & -3 & 2 \\ 0 & -2 & -1 & -3 \\ 0 & 0 & 2 & -1 \\ 0 & 0 & 0 & -4 \end{pmatrix}$$

$$\therefore \det(A) = (-2)(-2)(-2)(-4) = 32$$

Property 5, $\det$ unaltered

Step 1: $R_1 \leftrightarrow R_7$, $R_2 \leftrightarrow R_6$, $R_3 \leftrightarrow R_5$

Step 2: $R_1 \leftrightarrow R_4$, $R_2 \leftrightarrow R_3$

Step 3: $R_1 \leftrightarrow R_2$

N4.3.1

$$A = \begin{pmatrix} 6 & 9 & -6 \\ 0 & -8 & 4 \\ 2 & 5 & 0 \end{pmatrix}$$

$C_{11}(A)$

$$A_{11} = \begin{pmatrix} -8 & 4 \\ 5 & 0 \end{pmatrix}$$

$$\det(A_{11}) = (-8)(0) - (4)(5) = -20$$

$$C_{11}(A) = (-1)^{1+1}(-20) = -20$$

$C_{21}(A)$

$$A_{21} = \begin{pmatrix} 6 & -6 \\ 5 & 0 \end{pmatrix}$$

$$\det(A_{21}) = (6)(0) - (-6)(5) = 30$$

$$C_{21}(A) = (-1)^{2+1}(30) = -30$$

$C_{12}(A)$

$$A_{12} = \begin{pmatrix} 6 & 4 \\ 2 & 0 \end{pmatrix}$$

$$\det(A_{12}) = (6)(0) - (4)(2) = -8$$

$$C_{12}(A) = (-1)^{1+2}(-8) = 8$$

$C_{13}(A)$

$$A_{13} = \begin{pmatrix} 6 & -8 \\ 2 & 5 \end{pmatrix}$$

$$\det(A_{13}) = (6)(5) - (-8)(2) = 56$$

$$C_{13}(A) = (-1)^{1+3}(56) = 56$$

N4.3.1 continued

$$C_{22}(A) \quad A_{22} = \begin{pmatrix} 6 & -6 \\ 2 & 0 \end{pmatrix}$$

$$\det(A_{22}) = (6)(0) - (-6)(2) = 12$$

$$C_{22}(A) = (-1)^{2+2}(12) = 12$$

$$C_{23}(A) \quad A_{23} = \begin{pmatrix} 6 & -6 \\ 0 & -8 \end{pmatrix}$$

$$\det(A_{23}) = (6)(-8) - (-6)(0) = -48$$

$$C_{23}(A) = (-1)^{2+3}(-48) = 48$$

$$C_{31}(A) \quad A_{31} = \begin{pmatrix} 6 & 9 \\ 0 & -8 \end{pmatrix}$$

$$\det(A_{31}) = (6)(-8) - (9)(0) = -48$$

$$C_{31}(A) = (-1)^{3+1}(-48) = 48$$

$$C_{32}(A) \quad A_{32} = \begin{pmatrix} 6 & 9 \\ 2 & 5 \end{pmatrix}$$

$$\det(A_{32}) = (6)(5) - (9)(2) = 30 - 18 = 12$$

$$C_{32}(A) = (-1)^{3+2}(12) = -12$$

$$C_{33}(A) \quad A_{33} = \begin{pmatrix} 6 & -6 \\ 8 & 4 \end{pmatrix}$$

$$\det(A_{33}) = (6)(4) - (-6)(8) = 24 + 48 = 72$$

$$C_{33}(A) = (-1)^{3+3}(72) = 72$$

$$\therefore \det(A) = \begin{pmatrix} -20 & 8 & 56 \\ -30 & 12 & -12 \\ -48 & 48 & 72 \end{pmatrix}$$

$$\det(A) = \begin{pmatrix} -20 & 8 & 56 \\ 8 & 12 & -72 \\ 56 & -12 & -120 \end{pmatrix}$$

N4.41

$$A = \begin{pmatrix} 1 & 1 & 3 \\ 3 & 5 & 13 \\ 0 & 2 & 4 \\ 0 & 1 & -1 \\ 0 & 2 & -6 \\ 0 & 1 & 1 \end{pmatrix} \xrightarrow{\substack{R_2 - 3R_1 \\ R_4 - R_1 \\ R_5 - 2R_1 \\ R_6 - R_1}} \begin{pmatrix} 1 & 1 & 3 \\ 0 & 2 & 4 \\ 0 & 1 & -1 \\ 0 & 0 & 3 \\ 0 & 0 & -8 \\ 0 & 0 & -2 \end{pmatrix} \xrightarrow{\substack{R_3 \leftrightarrow R_2 \\ R_5 + 4R_2 \\ R_6 + 2R_2}} \begin{pmatrix} 1 & 1 & 3 \\ 0 & 1 & -1 \\ 0 & 2 & 4 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_3 - 2R_2} \begin{pmatrix} 1 & 1 & 3 \\ 0 & 1 & -1 \\ 0 & 0 & 6 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_4 - \frac{1}{2}R_3} \begin{pmatrix} 1 & 1 & 3 \\ 0 & 1 & -1 \\ 0 & 0 & 6 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 - R_2} \begin{pmatrix} 1 & 0 & 4 \\ 0 & 1 & -1 \\ 0 & 0 & 6 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 - 4R_3} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 6 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_2 + R_3} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 5 \\ 0 & 0 & 6 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_2 - 5R_3} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 6 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_3 \cdot \frac{1}{6}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

LU-Algorithm

$$\begin{aligned} -1 \ 1 \ 3 &\rightarrow /-1 = U_{r1} \\ 0 \ 2 \ 4 &\rightarrow /2 = U_{r2} \\ 0 \ 0 \ 3 &\rightarrow /3 = U_{r3} \\ 0 \ 0 \ 0 &\rightarrow /1 = U_{r4} \end{aligned}$$

Apply the U scalars along the diagonal, 1 if no pivot

Apply scalar and let pivot row leave unchanged

Enter the row op scalars that correspond

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -3 & 1 & 0 & 0 \\ 2 & -2 & 1 & 0 \\ 3 & 1 & -1 & 1 \end{pmatrix}$$

verifiable and elegant, I'm a fan

$$U = \begin{pmatrix} 1 & -1 & -3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

Exercises 3.1

1) a) $A = \begin{pmatrix} 2 & -1 \\ 3 & 2 \end{pmatrix}$

2x2 ad-bc
 $\det(A) = \frac{2}{4} \cdot \frac{2}{13} = 7$

b) $A = \begin{pmatrix} 6 & 9 \\ 8 & 12 \end{pmatrix}$

$\det(A) = \frac{6}{72} \cdot \frac{12}{72} = 0$

5) b) $A = \begin{pmatrix} 1 & -2 & 1 \\ -1 & 2 & 1 \\ 2 & -5 & 3 \\ 1 & 2 & 1 \\ -1 & 3 & 1 \\ 0 & 1 & 2 \\ 0 & 2 & 1 \end{pmatrix}$

$$\Rightarrow \begin{pmatrix} 1 & -2 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & -17 \end{pmatrix}$$

$\therefore \det(A) = 1 \times 1 \times (-17)$

Exercices 3.1

10) b)
$$\begin{pmatrix} 1 & 2 & 0 & 3 & 0 \\ -1 & 3 & 1 & 4 & 0 \\ 0 & 0 & 2 & 1 & 1 \\ 0 & 0 & -1 & 0 & 2 \\ 0 & 0 & 3 & 0 & 1 \end{pmatrix}$$

$$\det \begin{pmatrix} A & X \\ 0 & B \end{pmatrix} = (\det A)(\det B) \\ = 5 \times 7 \\ = 35$$

$$A = \begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix} \quad X = \begin{pmatrix} 0 & 3 & 0 \\ 1 & 4 & 0 \end{pmatrix}$$

$$0 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 2 & 1 & 1 \\ -1 & 0 & 2 \\ 3 & 0 & 1 \end{pmatrix}$$

$$\det(A) = \frac{1^3}{(1)(3)} / \frac{2^{+2}}{(2)(-1)} = 5$$

$$\det(B) = (2 \times 1 \times 1) + (1 \times 2 \times 3) + (1 \times (-1) \times 0) \\ - (1 \times 0 \times 3) - (2 \times 3 \times 0) - (1 \times (-1) \times 1) \\ = 7$$

Rule of
Sarrus

Drawing on Theorem 3.1.5

Exercices 3.2

22) b) find a polynomial $P(x)$ such that

$$p(0) = 5, \quad p(1) = 3, \quad p(2) = 5$$

$$p(0) = a(0)^2 + b(0) + c = c = 5$$

$$p(1) = a(1)^2 + b(1) + c = a + b + \frac{c}{5} = \frac{3-5}{-2}$$

$$p(2) = a(2)^2 + b(2) + c = 4a + 2b + \frac{c}{5} = \frac{5-0}{5}$$

$$\therefore b = -4$$

$$a + b = -2$$

$$4a + 2b = 0$$

$$(2a + b) - (a + b) = 0 - (-2)$$

$$\therefore a = 2$$

$$a = 2, \quad b = -4, \quad c = 5 \rightarrow p(x) = 2x^2 - 4x + 5$$

Exercises 3.2

Answer: $a = 7/6, b = 1/2, c = -1/3, d = 1$

$$\therefore p(x) = \frac{7}{6}x^3 + \frac{1}{2}x^2 - \frac{1}{3}x + 1$$

$$(a+c) + (-a-c)$$

$$= -\frac{1}{2} + \frac{1}{2}$$

23) b) Find a polynomial $p(x)$ of degree 3 such that

$$p(0) = p(1) = 1, p(-1) = 2, p(-2) = -3$$

$$p(0) = a(0)^3 + b(0)^2 + c(0) + d = 1$$

$$p(1) = a(1)^3 + b(1)^2 + c(1) + d = 0$$

$$p(-1) = a(-1)^3 + b(-1)^2 + c(-1) + d = 2$$

$$p(-2) = a(-2)^3 + b(-2)^2 + c(-2) + d = -3$$

$$\Rightarrow \begin{aligned} a+b+c &= 0 \\ -a+b-c &= 1/2 \\ -8a+4b-2c &= -1/2 \\ (a+b+c) + (-a+b-c) &= 0+1 \end{aligned}$$

$$(3a+c) - (a+b+c) = 3 - (-1/2)$$

$$2a = 7/2 \Rightarrow a = 7/4$$

$$7/6 + c = -1/2 - 7/6 = -2/3$$